A Shared accuracy model and latent-evidence reconstruction

Observed behavior

s = |orientation|, contrast c

Choice / correctness; high or low confidence

→

Fit accuracy first

$e = ks + \sigma_c \eta, \quad \eta \sim \mathcal{N}(0,1)$

Correct iff $e > 0$; fit k, σ_{lo}; fix $\sigma_{hi} = 1$

→

Reconstruct latent evidence

Sample $|e|$ conditional on s, c, and outcome.

4,000 draws per cell; fixed during confidence fitting.

Below, p is the high-confidence probability for a latent draw. Average over draws to predict the observed report.

B Confidence readouts, grouped by computation

Normalize decision strength

$|e|$ + noise scale → normalized strength

↔ Route 1, Fig. S4

↓

1 Calibrated · 2 parameters

$p = \Phi(\kappa|e|/\sigma_c + b)$

Adjust sensitivity and criterion.

Free: κ, b.

2 Bayesian(|e|) · 1 parameter

$p = \Phi(|e|/\sigma_c + b)$

Known contrast-specific noise; shift the confidence criterion.

Free: b (κ fixed at 1).

3 ContrastBlind · 2 parameters

$p = \Phi(\kappa|e|/\sigma_{pool} + b)$

One pooled noise scale at both contrasts.

Free: κ, b.

4 CASANDRE · 3 parameters

$p = \Phi(\kappa|e|/\hat{\sigma} + b)$

$\hat{\sigma} = \sigma_c \exp(\tau z), \quad z \sim \mathcal{N}(0,1)$

A noisy estimate of the noise scale.

Free: κ, b, τ.

Infer a posterior from scalar e

$e \rightarrow P(h|e), \quad h = (\text{sign}, s, c)$

↔ Route 2, Fig. S4

↓

5 FullBayes(e) · 1 parameter

$q = P(\text{chosen sign} \mid e), \quad p = \Phi(\Phi^{-1}(q) + b)$

Sum joint-posterior mass for the chosen sign; marginalize over magnitude and contrast; shift its criterion in probit space.

Free: b.

6 Entropy(e) · 2 parameters

$H^{-} = \sum_h P(h|e) \log P(h|e)$

$p = \Phi(\kappa H^{-} + b)$

Summarize concentration of the entire posterior;

do not select mass for the chosen sign.

Free: κ, b.

Both use the same joint hypothesis grid.
Flat priors over sign and 40 magnitude bins; empirical prior over contrast. The likelihood uses the parameters from the shared accuracy fit.

Use reconstructed chosen/unchosen pools

$(|e|, g_c) \rightarrow E^{+}, E^{-} \rightarrow p$

↔ Route 3, Fig. S4

↓

7 PositiveEvidence/Gain · 3 identified parameters

$(|e|, g_c) \rightarrow E^{+}, E^{-} \rightarrow p$

$E^{+} = (g_c + |e|)/2, \quad E^{-} = (g_c - |e|)/2$

$p = \Phi(\kappa[E^{+} - \alpha E^{-}] + b) = \Phi(Q|e| + B_c)$

Pools are reconstructed, not measured.

Nominal: κ, b, g_{lo}, α (fix g_{hi}=1); α=tanh(θ_α)∈(−1,1).

Only Q, B_{hi}, B_{lo} are identified (box below) — these 3 are what is fit and reported.

Only 3 combinations are identifiable:

$p = \Phi(Q|e| + B_c)$

$Q = \kappa(1+\alpha)/2$

$B_c = b + \kappa(1-\alpha)g_c/2$

κ, b, g_{lo}, α enter the likelihood only through this product — any α has an exactly compensating g_{lo} that reproduces the identical fit.

The data do not uniquely identify α (or g_{lo}) separately.

Reading the figure

Φ: standard normal CDF. κ: confidence sensitivity; b: confidence offset. $\sigma_{pool} = \sqrt{((\sigma_{hi}^2 + \sigma_{lo}^2)/2)}$.

Counts refer to confidence parameters beyond the shared accuracy fit. Every model here has at least one free confidence parameter (bare zero-parameter ideal-observer versions of Bayesian(|e|) and FullBayes(e) were considered but are not part of the reported model set).

Source: manuscript Methods (free-parameter models of real confidence reports) and Supplemental Information (identifiability discussion).